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RESEARCH ARTICLE

Performance of a tractor-mounted probe for undisturbed soil sampling: implications for soil organic carbon stocks

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Summary

The assessment of soil organic carbon (SOC) stocks relies on several key factors, including the total SOC content of the soil, the bulk density (BD) of the soil, and the depth of the sampled layers. However, traditional methods, particularly those using volumetric cylinders for undisturbed soil sampling, present significant logistical challenges for large-scale projects due to their costly and time-consuming nature, often requiring the excavation of trenches. In response, automated probes, which are commonly used in geological studies, offer a promising alternative, but their application in the context of BD soil studies remains under discussion. The objective of this study was to evaluate the efficacy of a tractor-mounted probe in comparison to the conventional core method for the collection of undisturbed soil samples across a range of soil textures and depths. The results indicated that the coefficient of variation (CV) for the probe's bulk density (BD) measurements ranged from 3 to 15% in sandy soils but remained consistently below 5% in clay-sandy-loam and clay textures. Despite small differences in BD values between the methods used, SOC stock assessments showed only minor variations across all layers, regardless of soil texture. Therefore, this study demonstrated that the tractor-mounted probe represents a viable and scalable solution for conducting large-scale SOC stock assessments, despite its susceptibility to minor variations in sandy soil conditions. This research contributed to the field of automated soil sampling tool validation and offered alternatives for field operationalization in large-scale carbon projects. It also raised new questions for further investigation into other field conditions

Keywords: Soil bulk density; soil sampling methods; automatized soil sampling; carbon projects

Introduction

In the constantly evolving scenario of greenhouse gas emissions, the soil has a fundamental role as an atmospheric CO_2 sink and carbon storage in the biosphere (Davidson and Janssens, 2006; Lal, 2013; Smith, 2016). The total soil organic carbon (SOC) ranges from 504 to 3000 PgC up to the first 100 cm depth (Scharlemann et al., 2014). It also represents approximately three times the stocks of carbon (C) found in vegetation and twice the stocks of C in the atmosphere (Smith, 2012). Furthermore, in the context of climate change, protocols are being developed worldwide to measure, monitor, report, and verify SOC in agricultural landscapes (FAO 2019). To meet the

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global demand for SOC inventory, the IPCC Guidelines for National Greenhouse Gas Inventories provide a comprehensive sampling protocol (IPCC, 2019).

In the context of SOC quantification, the soil bulk density (BD) is used to calculate SOC stock, in Mg C ha^{-1}, for a specified soil layer by considering the soil's mass-bulk volume relationship. The importance of BD accuracy for precise estimation of SOC stocks is well established in the literature (Lal and Kimble, 2000; Cihacek et al., 2015; Walter et al., 2016). These authors assert that the accuracy of SOC stock estimates is not solely dependent on the use of modern and accurate SOC quantification methods. If there are errors in the sampled BD to the actual field BD, all data will be compromised.

The soil core method is the most commonly used method for obtaining BD. This method uses a cylinder of known volume to measure and quantify the soil within the sampled bulk volume. This is the most accurate method, despite the existence of several indirect methods, which will not be detailed in this work (e.g., radiation, remote sensing, X-ray, seismic waves) (Blake and Hartge, 1986; Aitkenhead and Coull, 2020; Romero-Ruiz et al., 2021). Extracting a core from the soil requires various instruments and tools, with the soil core sampler being the most widely used. This tool comprises a 'housing' that holds the cylinder, which is driven into the soil using a hammer or applying pressure (Erbach, 1987; FAO, 2019; Parfitt et al., 2010; Walter et al., 2016).

Regardless of the tools used, inserting the cylinder into the soil experiences a series of physical and mechanical phenomena. For example, compaction and soil loss are some mechanical barriers to collecting a suitable sample (Blake and Hartge, 1986; Walter et al., 2016; Throop et al., 2012). The impact of these factors on sample quality is determined by, e.g., soil moisture, the number of blows, and the force of hammering. Measuring SOC stocks at depths of 100 cm, for example, often requires digging large trenches, making the process labour-intensive, costly, and time-consuming. Consequently, this method is technically and economically unfeasible for large-scale carbon projects.

Alternatively, vehicle-mounted probes have been historically developed for direct soil sampling without opening the trench (Robertson et al., 1974; Vepraskas et al., 1990; Leonov and Sankina, 2020; Olmedo et al., 2020). These probes aim to increase the number of collections in a short space of time, reducing soil disturbance and the need for trenches. Also, by automating the process, they significantly reduce human dependency. However, the reliability of new probes relative to traditional methods needs to be scientifically proved (Walter et al., 2016); otherwise, their use in carbon projects won't be possible.

In this context, the study was based on the hypothesis that automated probes can produce similar BD values as traditional methods (in this study, the core method) ensuring that SOC stocks remain unchanged by the soil sampling method chosen. The objective of the study was to evaluate the performance of a tractor-mounted probe for BD sampling and its implications on SOC stocks in three types of soils with different textures.

Materials and methods

Study sites and soil characterization

The study was conducted in the Tangara da Serra region of the State of Mato Grosso, Brazil. Three experimental sites were selected based on particle size distribution and homogeneity. They are named sandy (14°22′02′′S, 57°33′53′W), medium (15°07′07.8′′S, 56°56′33.1′′W), and clayey (14°37′00.0′′S, 57°31′15.4′′W). Each site has the following clay, silt, and sand contents: 7, 1, and 92% (Sand), 25, 8, and 67% (Sandy Clay Loam), and 50, 5, and 45% (Clay), respectively, and all three sites were covered with *Brachiaria ruziziensis* pasture

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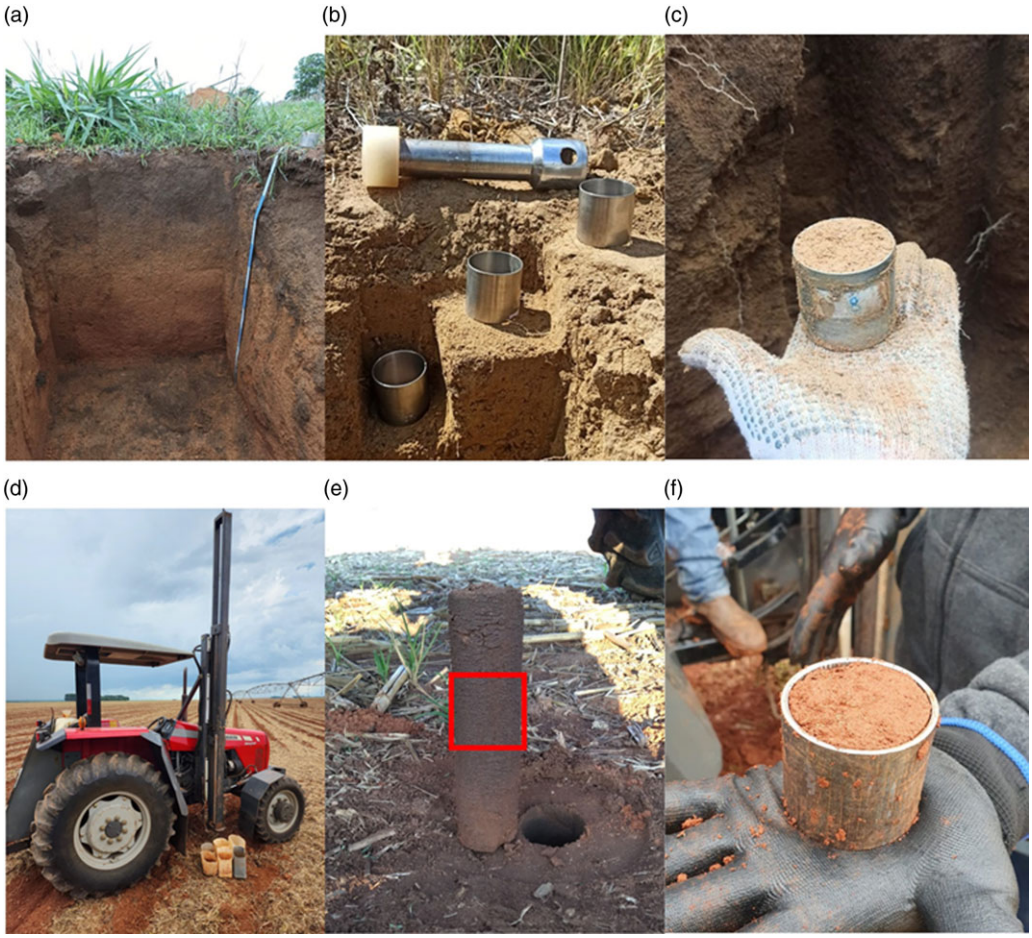

Figure 1. The illustrative procedure of the standard method soil sample collection (soil core method): (a) opened trench; (b) standard core sampler used, and (c) sample collected; and the soil sampling using tractor-mounted probe: (d) equipment used; (e) sample collected; and (f) final sample.

Soil sampling

Three trenches (one at each site) of approximately 1 m^2 and 150 cm depth were dug in each soil (Figure 1a). At each site, six samples of undisturbed soil cores were collected vertically at depths of 0–10, 10–20, 20–30, 30–40, 40–60, 60–80, and 80–100 cm on the sides of the trench using both the manual core method and the tractor-mounted probe (Figure 2). Additionally, two disturbed soil samples were collected from the trench wall adjacent to the samples taken using the standard method, to determine the SOC and water content. The average of these two samples was calculated to establish the moisture and SOC content. Figure 2 shows the spatial arrangement of the undisturbed soil sampling on the soil surface. Sampling points were separated by 50 cm to avoid soil disturbance.

The soil characterization for volumetric water (θ) and SOC content of the sandy, medium and clayey experimental soil sites is presented in Table 1. It is important to note that only the sandy site was significantly moist, which facilitated the soil sampling in this area.

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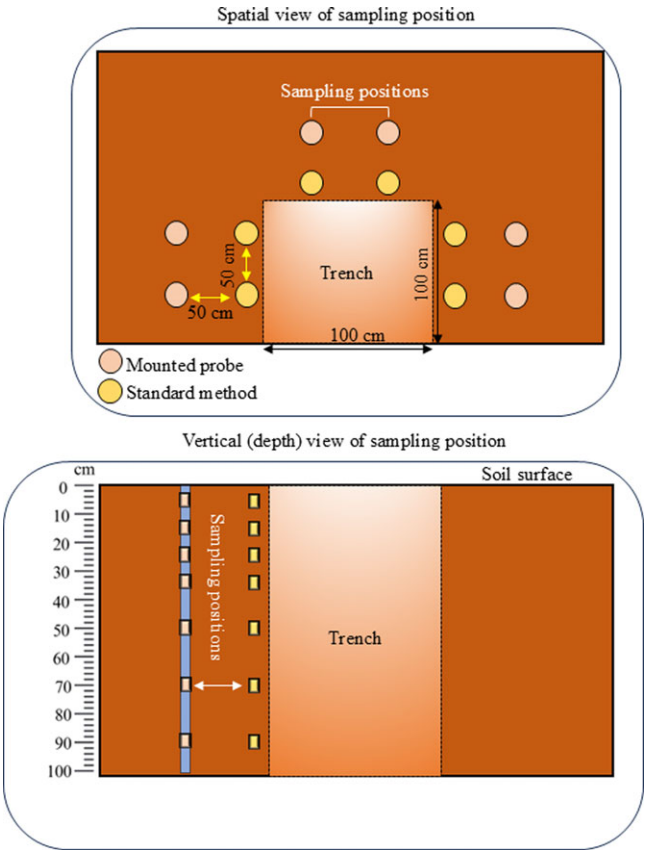

Figure 2. Illustrative scheme of the superficial (upper) and vertical (bottom) spatialization of sampling position by the tractor-mounted probe and standard method (soil core method).

Description of the sampling methods

The performance of the tractor-mounted probe for undisturbed soil sampling was evaluated in contrast to the standard method. The standard method was a 100 cm^3 sample cylinder (5 cm diameter and 5 cm height) collected using a hammer and a sampler, as described by Walter et al. (2016). This cylinder size is widely and frequently employed for BD sampling (ISO, 2017). The standard method involved digging a 100 cm deep trench (Figure 1a) and extracting soil cores from the middle portion of each layer using a hammered core soil sampler (Figure 1b) to obtain the standard soil sample (Figure 1c).

A tractor-mounted probe (patent registered in the National Institute of Industrial Property – INPI, process number: BR 2020230145480, Supplemental Figures 1–3) with a hydraulic system was used to extract samples from a single hole with a speed of 5 cm per second. The probe, a bevelled metal tube measuring 0.0498 metres in diameter and 1 metre in length, was continuously introduced into the soil until it reached the predefined depth (see section above) (Figure 1d). Once the desired sampling depth is reached, a vacuum in the sample tube holds a 10 cm sample in place. As the probe was withdrawn from the soil, the tractor's hydraulic device forced the soil sample out of the tube (Figure 1e). To standardize the sampling process and ensure uniform sample size, a cylindrical structure with the same diameter as the probe and a height of 5 centimetres was used. The sampled soil structure was then incrementally surrounded with a 5 cm diameter metal cylinder. The surplus material on both sides (top and bottom) of the sample (2.5 centimetres on

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Table 1. Volumetric water (θ) and soil organic carbon (SOC) content of the sandy, medium, and clayey experimental soil sites

	Sandy*	Medium	Clayey	Sandy	Medium	Clayey
Layers (cm)		θ (m^3 m^{-3})			SOC (g kg^{-1})	
0–10	0.17	0.03	0.13	9.89	3.56	18.40
10–20	0.14	0.04	0.13	8.48	3.67	13.40
20–30	0.11	0.05	0.13	6.60	3.94	11.40
30–40	0.10	0.06	0.13	6.27	4.13	9.67
40–60	0.12	0.05	0.13	6.76	6.24	9.40
60–80	0.12	0.06	0.13	5.39	3.32	9.02
80–100	0.11	0.06	0.14	4.23	2.23	7.68

*Samples were collected after rain in the area.

each side) was carefully removed, resulting in the retention of the central portion of the sampled layer (Figure 1f). The middle section (5 cm in height) of the entire (10 cm in height) sample was used as the final core sample.

Soil analysis

The undisturbed soil samples were weighed to determine the wet soil mass, followed by oven-drying at 105°C for 48 hours. The soil BD was calculated as the ratio between the weight of the oven-dried soil and the total volume of the cores. The gravimetric water content (w) was determined by the ratio of water to the mass of oven-dried soil. To determine the total C content, the disturbed soil samples were sieved at 2 mm mesh and then oven-dried at 45°C until constant soil mass was reached. After drying, the samples were ground and sieved at 0,150 mm mesh for C determination by the dry combustion method (Nelson and Sommer, 1982), using an Elemental Analyzer – LECO/Truspec. SOC stocks were then calculated using Eq. 1, with BD and C content values as inputs.

$$Ce = \frac{C\ x\ BD\ x\ L}{10} \tag{1}$$

where C_e (Mg C ha^{-1}) is the SOC stock; C is the total organic carbon (g kg^{-1}); BD is the soil bulk density (Mg m^{-3}); and L is the thickness of the soil layer thickness (cm).

Data analyses

Performance comparisons of the tractor-mounted probe in relation to the soil standard soil sampling method were performed by layer, i.e., comparisons between layers. The data were subjected to an analysis of variance (ANOVA), and the means were compared using a 95% confidence interval. Furthermore, the coefficient of variation (CV%), relative deviation, and absolute differences between means were employed to assess data variability. SOC stocks for 0–30 and 0–100 cm, resulting from samples collected by the standard and tractor-mounted probe methods, were analysed using an F-test at the $\alpha = 0.05$ significance level.

Results and discussion

The BD measurements calculated for each soil at different depths clearly showed that the dispersion of BD obtained by the tractor-mounted probe was significantly greater than those found using the standard method for the sandy soil. The standard method had a CV of less than 5%, while the tractor-mounted probe yielded values between 3% and 15% (Figure 3). However,

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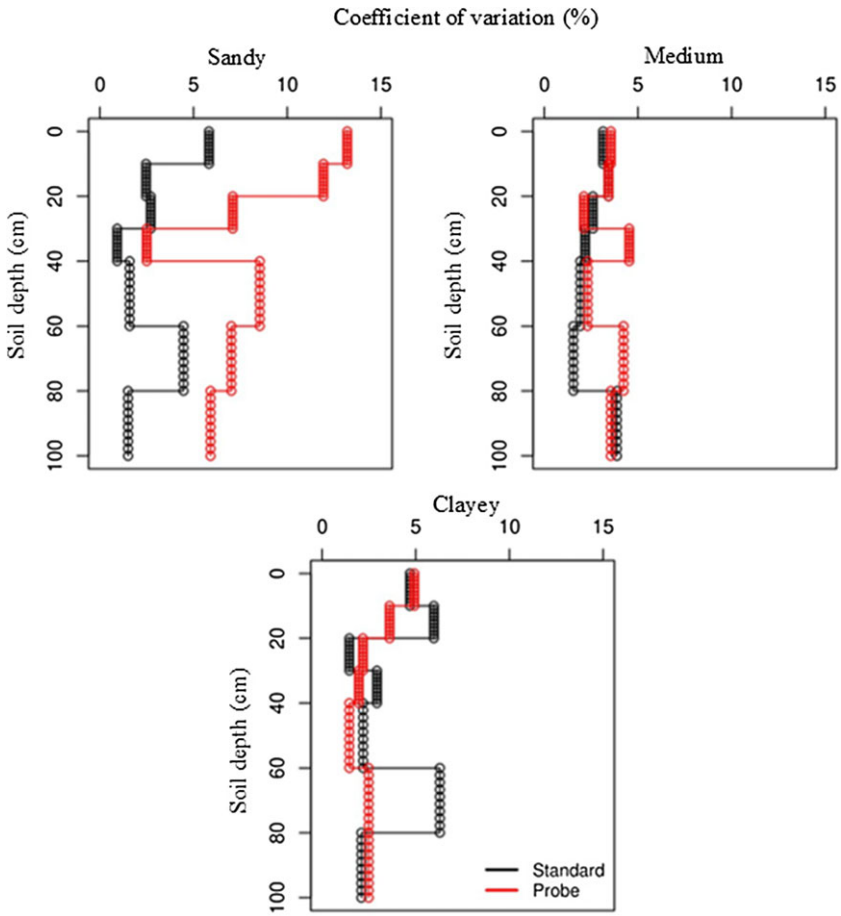

Figure 3. Coefficient of variation (CV) of BD ($n = 6$) obtained for the tractor-mounted probe and the soil core standard method for sandy, medium, and clayey-textured soils at different soil layers.

there are clear similarities in the BD variability obtained by the tractor-mounted probe and the standard method for the medium and clayey soils. Both methods yielded CV values below 5% (Figure 3).

Only punctual differences (non-overlapping confidence intervals) were observed between the BD means of the tractor-mounted probe and the standard method for the three soils (Figure 4). These differences were found in the layers with the lowest CV values, specifically in the 30–40 cm sandy layer, and the 20–30 cm and 80–100 cm clayey layers. The largest confidence intervals were found for sandy soil, with the average BD ranging from 1.37 to 1.56 Mg m^3 for the tractor-mounted probe in the surface layer (0–10 cm), and from 1.44 to 1.55 Mg m^3 at the 40–60 cm layer. Confidence intervals were reduced as the medium and clayey soils were examined (reflecting a lower CV). Higher BD values in the clayey texture at surface layers in comparison to deep ones, might be due to compaction in the area caused by the use of agricultural machinery over the years.

It is important to note that the means for C stock calculated from the BD obtained by the methods are practically the same (Figure 5), leading to overlap and reduced confidence intervals for the means. This makes it difficult to detect any statistical difference (Figures 4 and 5). The SOC stock obtained by the tractor-mounted probe and the standard method is practically the same for the medium soil (overlap and reduced confidence intervals). A similar overlap in confidence

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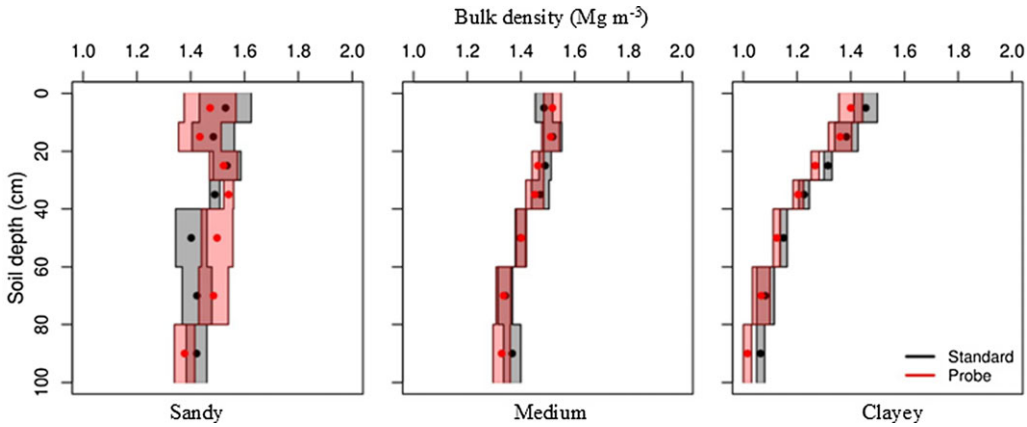

Figure 4. Means and confidence interval for the means (95%) of BD obtained for the tractor-mounted probe and the soil core standard method for sandy, medium, and clayey-textured soils at different soil layers.

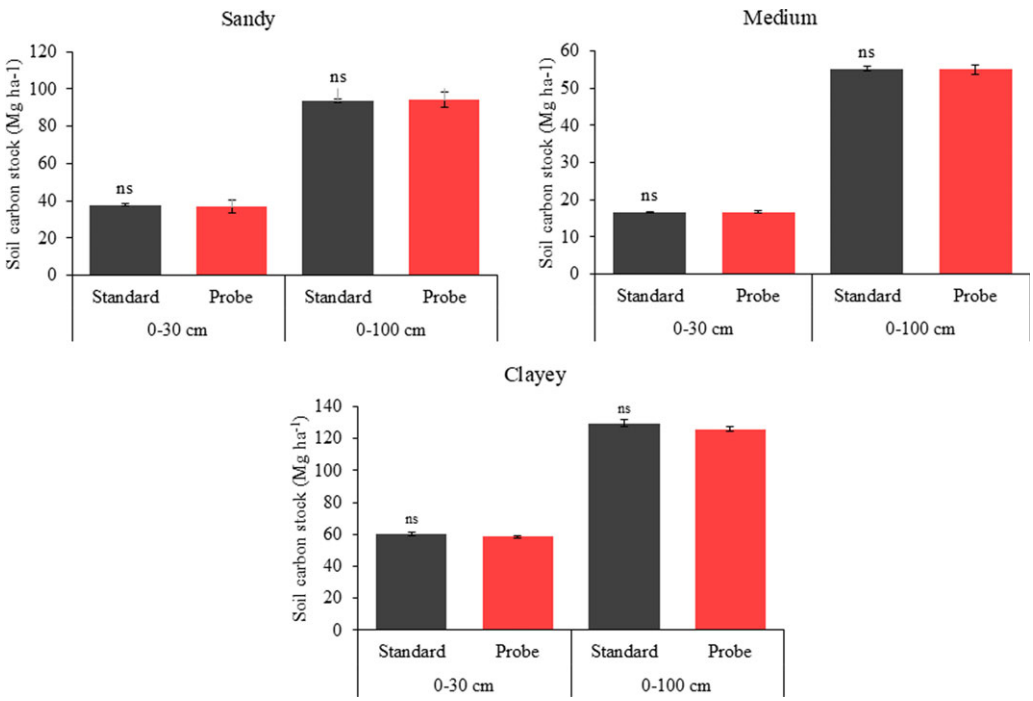

Figure 5. Comparison of the SOC stocks for 0–30 and 0–100 cm layers obtained by the standard method and the tractor-mounted probe across sandy, medium, and clay soil textures. Ns: non-significant according to the F-test conducted at the $\alpha = 0.05$ significance level.

intervals can also be observed for clayey soil. Furthermore, the cumulative SOC stocks for the 0–30 and 0–100 cm depths (Figure 5) remained unchanged when comparing the standard method and tractor-mounted probe across all soil textures examined.

Our findings also revealed that even without a statistical difference in BD between the two methods in the sandy soil, absolute differences between the methods reached ~ 0.1 Mg m^{-3} at around 40–80 cm, whereas the lowest differences (~0.03 Mg m^{-3} and ~0.05 Mg m^{-3}) were

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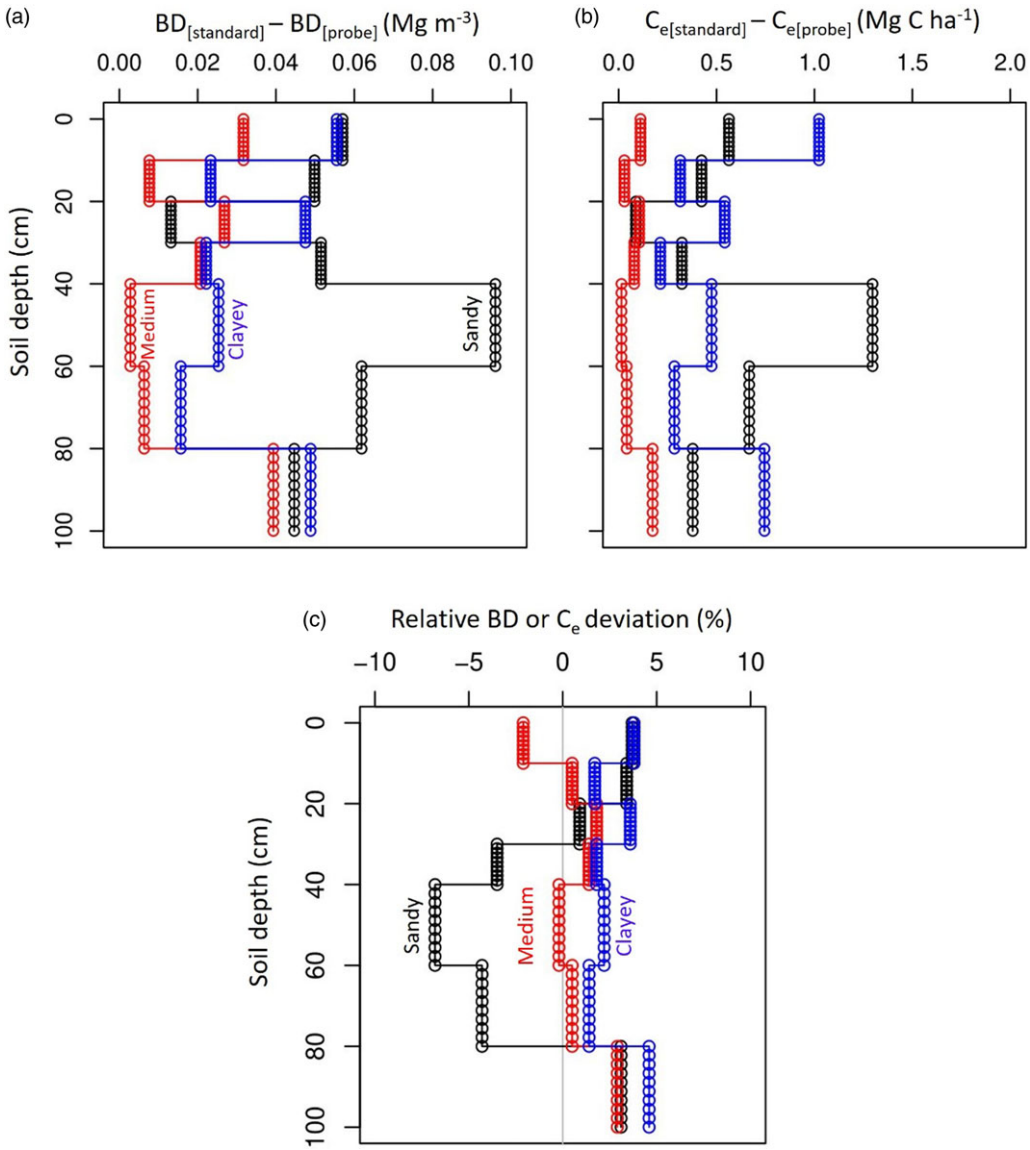

Figure 6. Absolute differences between means of BD ($BD_{[standard]}$ – $BD_{[probe]}$) and SOC stocks ($C_{e[standard]}$ – $C_{e[probe]}$) obtained for the tractor-mounted probe and the soil core standard method for sandy, medium, and clayey-textured soils at different soil layers (a, b). Relative deviation of the tractor-mounted probe in relation to the standard method for BD or SOC stock across sandy, medium, and clay soil textures.

obtained for the medium and clayey soil (Figure 6a). The absolute differences between the methods for SOC stocks in the sandy soil reached a maximum of 1.3 Mg C ha^{-1} (40–60 cm), while this difference was close to zero for the medium soil (Figure 6b). In the clayey soil, the difference was 1.0 Mg C ha^{-1} in the surface layer (0–10 cm) and lower differences in depth.

Figure 6c indicates that the relative differences between the standard method and the tractor-mounted probe reached approximately 5% for sandy soil, i.e., BD or SOC stocks obtained by the tractor-mounted probe were 5% lower (40–60 cm) or 4% higher (10–20 cm) in this soil. For the medium soil, the values were no greater or less than ± 3%, but with differences very close to zero (40–60 cm) or 1% (60–80 cm). The tractor-mounted probe yielded values that were 2–4% greater

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than the standard method for the clayey soil, with variations across layers. When all layers are considered, the differences between the standard method and the tractor-mounted probe for sandy, medium, and clayey soils are 3.6, 1.3, and 2.7%, respectively.

Performance of the tractor-mounted probe in the soil textures

The comparison between a tractor-mounted probe and a 100 cm^{-3} cylinder (taken as a standard method) to determine BD and SOC stocks produced average differences that did not exceed 5%, with greater variability peaks for the sandy soil. Previous studies comparing alternative methods with the standard method also found that deviation peaks depend on the soil type (Parfitt et al., 2010; Cihacek et al., 2015; Walter et al., 2016). It is important to note, however, that certain factors, such as the power of the hammer and the intensity with which the probe is inserted, the shape of the cutting shoe, and the friction on the soil-cylinder wall, significantly influence the variation observed between methods (Walter et al., 2016; Lima and Keller, 2019).

The sandy soil showed the highest variability and the greatest absolute differences between the tractor-mounted probe and the standard method. As Walter et al. (2016) correctly point out, this higher variability is due to the low structural stability of sandy soils. The tractor-mounted probe is designed to continuously and precisely introduce the probe into the soil and extract the sample to the desired depth. The reduced natural cohesion of sandy soils (Hillel, 2003) can make it difficult to achieve a defined soil mass within the core, causing higher variability. Furthermore, Walter et al. (2016) reported stretching during sample extraction, which is more likely to occur in soils with low cohesion, such as sandy soils. In this context, sandy soils present a significant challenge when not in high moisture conditions, which would require adjustments to the tractor-mounted probe. These adjustments would ensure the complete sampling of the soil.

The percentage deviations in BD obtained by the tractor-mounted probe in relation to the standard method were found to range from negative (in sandy soil) to positive (in clayey soil). The tractor-mounted probe does appear to slightly compact soils with higher clay content, although these deviations are not greater than 3%. Throop et al. (2012) assert that compaction is more evident in cohesive soils than in coarse soils. In clayey soil, the average deviation of 2.5% was positive throughout the soil profile. Clay soils exhibit slightly plastic behaviour, which is dependent on soil moisture (Keller and Dexter, 2012; Schjønning et al., 2023). We did not examine water content as a factor of experimental variation, but we can confirm that the soil was slightly moist at the time of sampling. Any excessive force applied to insert the ring into the soil will cause compaction inside the ring, whether using the standard method or the tractor-mounted probe (Lima and Keller, 2019). These experiments were conducted under moisture conditions that allowed us to verify that compaction was slightly more evident in the tractor-mounted probe method. However, the deviations produced with compaction were minimal, with peaks of no more than 4%. The medium-textured soil showed no signs of compaction.

Impact of BD differences on SOC stock calculations

In this study, the same values of soil organic carbon contents were used in the calculation of SOC stock in both BD sampling methods. This allowed us to incorporate all BD variability into the SOC stock calculation. As reported by Lal and Kimble (2000), Cihacek et al. (2015), and Walter et al. (2016), the systematic error in BD with the tractor-mounted probe sampling will cause an error propagation in the estimated SOC stocks. Our results definitively showed that these errors were greater for sandy soils. However, given that some studies have reported coefficients of variation between 10 and 17% for SOC stocks using a fixed method for

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undisturbed soil sampling (e.g., Cihacek et al., 2015; Walter et al., 2016; Santos et al., 2019), it is clear that the differences produced between the tractor-mounted probe and the standard method fall within an acceptable CV range (Figure 2). This demonstrates that, regardless of the method used, some variability is introduced (Walter et al., 2016). The small differences observed of 0.5–1.0 Mg C ha^{-1} per layer (Figure 6b) for SOC stocks found in this study between sampling methods are relatively low. It can be inferred that, if the sampling method were standardized using the probe from start to finish, the results would likely be very close to those obtained by the standard method for carbon inventories.

There were no significant changes in the SOC stocks between the two tested methods. Cihacek et al. (2015) definitively found no significant deviations when examining BD for SOC calculation using sampling tubes. This clearly shows that the differences in the methods result in similar averages for both BD and SOC stocks. However, refinements to the tractor-mounted probe will undoubtedly enhance its accuracy. For example, in cases where uncertainties are observed with a particular method (standard or alternative), increasing the number of replications will lead to more reliable averages. Before making any adjustments to enhance the collection mechanics of the tractor-mounted probe, we must first consider increasing the number of samples, mainly for the sandy soil and evaluating different soil textures with a range of moisture conditions.

Conclusions

The comparison between a hydraulic tractor-mounted probe and the standard method (i.e., manual collection of cylinders in the trench wall) to determine BD resulted in average differences that did not exceed 5%, with greater variability peaks observed in sandy soil. Absolute differences in BD between the standard method and the tractor-mounted probe for sandy, medium and clayey soils were 3.6, 1.3, and 2.7 %, respectively. Our study suggests that BD values obtained in the soil sampling using the tractor-mounted probe did not insert extra uncertainties in the SOC stock calculations compared to the standard method. There was no change in SOC stocks between the two sampling methods for the 0–30 and 0–100 cm layers. Hence, the tractor-mounted probe represents a viable and agile alternative for the collection of undisturbed samples in large-scale soil health and carbon projects, particularly in clay and structured soil types. In areas characterized by limited accessibility, such as mountainous terrains, steep slopes, or densely forested regions, the application of a tractor-mounted probe is rendered impractical due to the inability of the tractor to traverse these challenging environments. Furthermore, the physical exertion necessary to manoeuvre the probe in such conditions renders it impractical, thereby making standard methods more suitable for soil sampling in these regions. Finally, we recommend that the tractor-mounted probe should be tested in other soil conditions such as tilled soils and soils with different moisture levels.

Supplementary material. The supplementary material for this article can be found at https://doi.org/10.1017/S0014479725000031

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Competing interests. All authors declare that they have no conflicts of interest.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work, the author(s) used DeepL Write and ChatGPT in order to improve the English language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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